Out of Distribution Generalization: Invariant Risk Minimization and Risk Exploration

Behrad Moniri

Department of Electrical Engineering
Sharif University of Technology

May 25, 2020

Out-of-Distribution Generalization via Risk Extrapolation

David Krueger [1 2] Ethan Caballero [1] Jörn-Henrik Jacobsen [3 4] Amy Zhang [1 5 6] Jonathan Binas [1 2]
Rémi Le Priol [1 2] Aaron Courville [1 2]

ICLR 2020: Workshop on Causal Learning For Decision Making

Introduction

Out-of-Distribution (OoD) Generalization

A model must generalize to new distributions at test time without seeing any training data from them.

Introduction

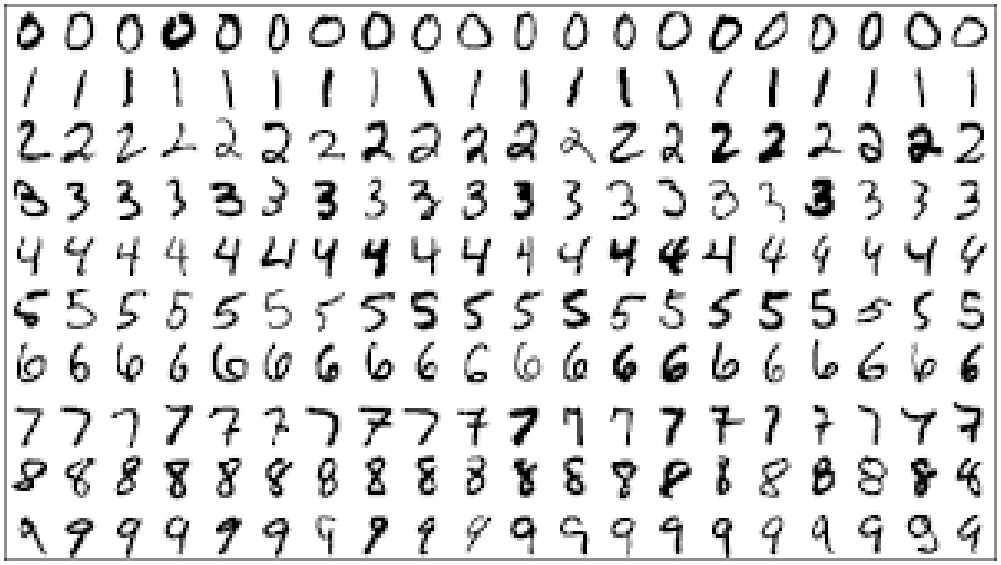
Out-of-Distribution (OoD) Generalization

A model must generalize to new distributions at test time without seeing any training data from them.

Assumptions

We assume a fixed underlying task, and access to labeled data from multiple training environments. We also assume that variation in these environments is somewhat representative of the variations we will see at test time.

Colored MNIST



Digits are colored either red or green.
Train Env 1: 80% Correlation with labels.
Train Env 2: 90% Correlation with labels.
Test Env: 80% Correlation with labels.

Other Examples

Other Examples

Structural Causal Model

Consider the following structural causal model:

$$\begin{cases} X_1 = \text{Gaussian}(0, \sigma^2) \\ Y = X_1 + \text{Gaussian}(0, \sigma^2) \\ X_2 = Y + \text{Gaussian}(0, 1) \end{cases}$$

Three kinds of regressions can be performed Only one is invariant with σ.
Question: Can the invariant explanation generalize?

Notations and Basics

- $\mathcal{E}_{tr}$: The set of environments available in training.
- $\mathcal{E}_{all}$: The set of environments.
- From each environment e, i.i.d. sample are drawn from $\mathcal{D}_e$.
- Risk function of the estimator f_θ for environment e:

$$R_e(\theta) = \mathbb{E}_{(X,Y)\sim\mathcal{D}_e}\Big[l(f_\theta(x), y)\Big]$$

Methods

- **Empirical Risk Minimization** [Vapnik, Chervonenkis, Kearns, Valiant]:

$$R_{ERM}(\theta) = \mathbb{E}\Big[l(f_\theta(x), y)\Big]$$

Methods

- **Empirical Risk Minimization** [Vapnik, Chervonenkis, Kearns, Valiant]:

$$R_{ERM}(\theta) = \mathbb{E}\Big[l(f_\theta(x), y)\Big]$$

- **Robust Optimization** [Ben-Tal et al., 2009]: Minimize the error in the worst environment!

$$R_{RO}(\theta) = \max_{e \in \mathcal{E}} \Big[R_e(\theta) - r_e\Big]$$

Invariant Risk Minimization

Philosophical Point: Why is gravity so important?

Invariant Risk Minimization

Philosophical Point: Why is gravity so important?
Idea: an invariant representation $\Phi(X)$ is one such that the optimal linear predictor ω is the same across all environments e.

Invariant Risk Minimization

Philosophical Point: Why is gravity so important?
Idea: an invariant representation $\Phi(X)$ is one such that the optimal linear predictor ω is the same across all environments e.

$$\min_{\Phi,\omega} \sum_{e\in\mathcal{E}} R_e\big(\omega^T \Phi(X^e)\big)$$
$$s.t.\ \omega \in \operatorname*{argmin}_{\bar{\omega}} R_e\big(\bar{\omega}^T \Phi(X^e)\big) \ \ \forall e \in \mathcal{E}$$

This optimization problem is very hard:

$$\min_{\Phi} \sum_{e\in\mathcal{E}} R_e\big(\Phi(X^e)\big) + \lambda \Big|\Big|\nabla_w R_e(\omega^T \Phi(X^e))\Big|\Big|_2^2$$

Minimax Risk Exploration

Our Goal:

- Reduce training risks.
- Increasing similarity of training risks.

Minimax Risk Exploration

Our Goal:

- Reduce training risks.
- Increasing similarity of training risks.

This paper introduces **Minimax REx** and **Variance REx**, for achieving, and balancing these goals.

Minimax Risk Exploration

Our Goal:

- Reduce training risks.
- Increasing similarity of training risks.

This paper introduces **Minimax REx** and **Variance REx**, for achieving, and balancing these goals. **Risk Interpolation:**

$$R_{RI}(\theta) = \max_{\sum_{e\in\mathcal{E}} \lambda_e = 1,\ \lambda_e \geq 0} \sum_{e\in\mathcal{E}} \lambda_e R_e(\theta)$$

Minimax Risk Exploration

$$R_{RI}(\theta) = \max_{\sum_{e\in\mathcal{E}} \lambda_e = 1,\ \lambda_e \geq 0} \sum_{e\in\mathcal{E}} \lambda_e R_e(\theta)$$

Minimax Risk Exploration

$$R_{RI}(\theta) = \max_{\sum_{e\in\mathcal{E}} \lambda_e = 1,\ \lambda_e \geq 0} \sum_{e\in\mathcal{E}} \lambda_e R_e(\theta)$$

Minimax Risk Exploration:
We relax this optimization problem.

$$R_{MM-REx}(\theta) = \max_{\sum_{e\in\mathcal{E}} \lambda_e = 1,\ \lambda_e \geq -\beta} \sum_{e\in\mathcal{E}} \lambda_e R_e(\theta)$$

We can easily solve this linear programming problem:

$$R_{MM-REx}(\theta) = (1 + m\beta) \max_{e\in\mathcal{E}} R_e(\theta) - \beta \sum_{e=1}^{m} R_e(\theta)$$

Minimax Risk Exploration

$$R_{RI}(\theta) = \max_{\sum_{e\in\mathcal{E}} \lambda_e = 1,\ \lambda_e \geq 0} \sum_{e\in\mathcal{E}} \lambda_e R_e(\theta)$$

Minimax Risk Exploration:
We relax this optimization problem.

$$R_{MM-REx}(\theta) = \max_{\sum_{e\in\mathcal{E}} \lambda_e = 1,\ \lambda_e \geq -\beta} \sum_{e\in\mathcal{E}} \lambda_e R_e(\theta)$$

We can easily solve this linear programming problem:

$$R_{MM-REx}(\theta) = (1 + m\beta) \max_{e\in\mathcal{E}} R_e(\theta) - \beta \sum_{e=1}^{m} R_e(\theta)$$

- $\beta = 0$: Risk Interpolation.
- $\beta = \frac{-1}{m}$: Empirical Risk Minimization.

Regularized Optimization and V-REx

A simpler approach that we find works well in practice:

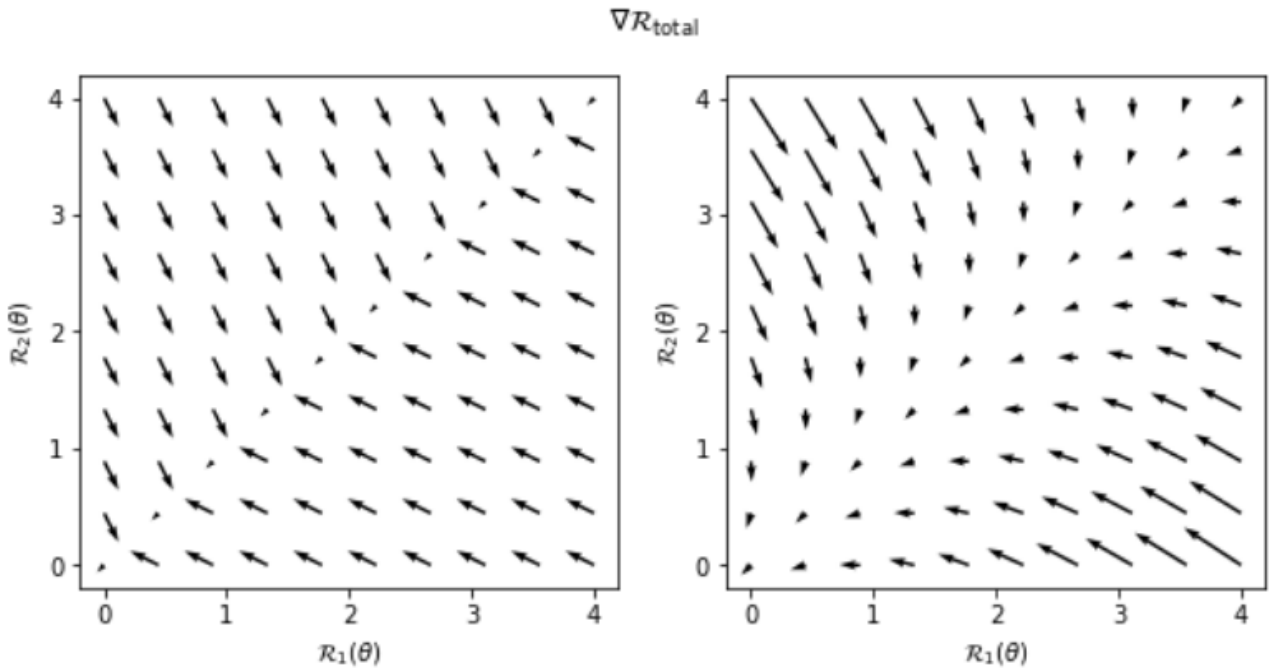
$$R_{V-REx} = \beta \mathrm{Var}[R_1, \ldots, R_m] + \sum_{e=1}^{m} R_e$$

Regularized Optimization and V-REx

A simpler approach that we find works well in practice:

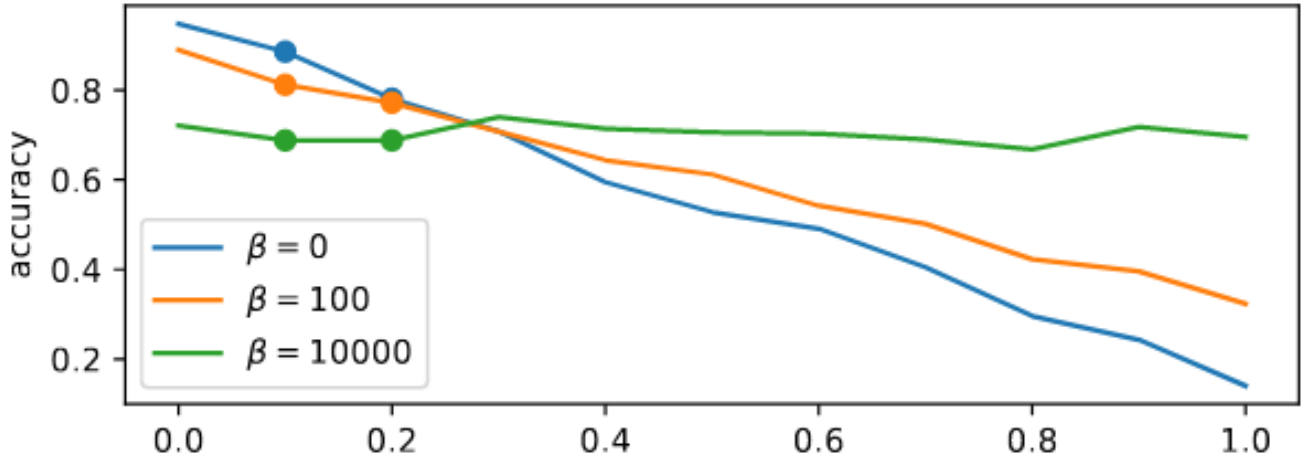
$$R_{V-REx} = \beta \mathrm{Var}[R_1, \ldots, R_m] + \sum_{e=1}^{m} R_e$$

Left: **REx**, Right: **V-REx**:



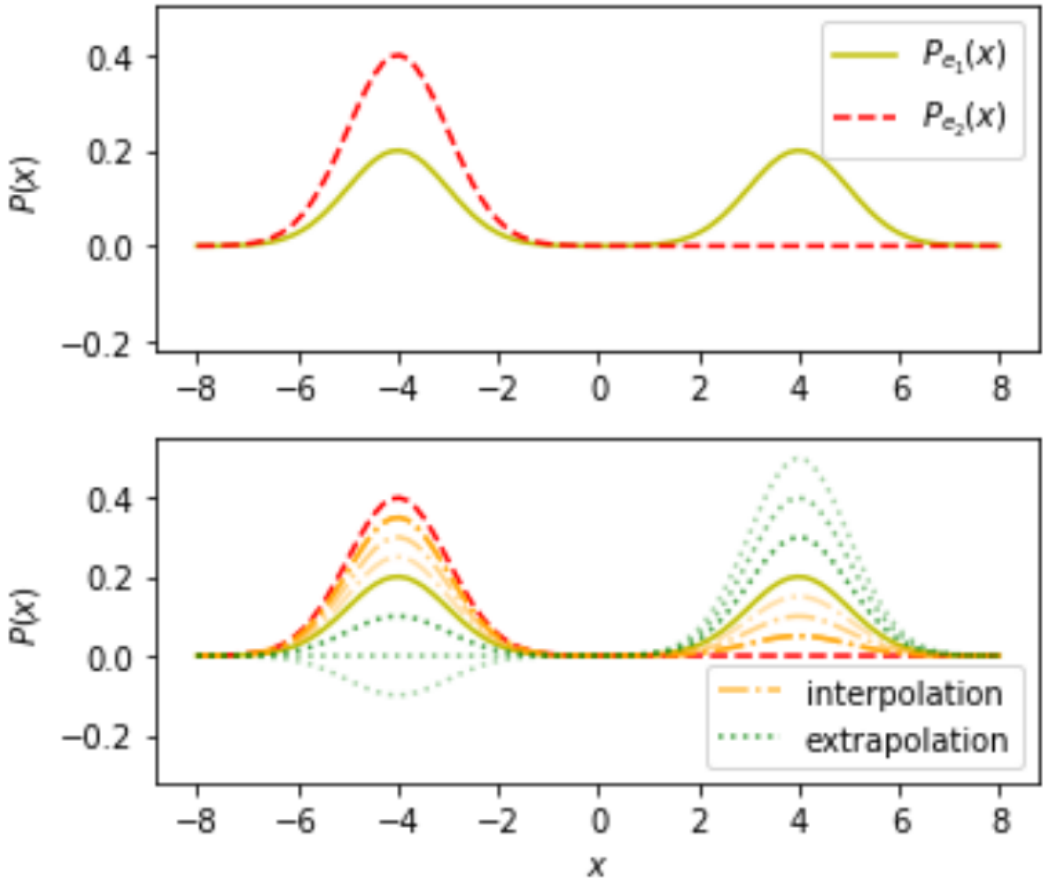
Color MNIST and V-REx

$$R_{V-REx} = \beta \mathrm{Var}[R_1, \ldots, R_m] + \sum_{e=1}^{m} R_e$$



Probabilistic View :((

$$R_{MM-REx}(\theta) = \max_{\sum_{e\in\mathcal{E}} \lambda_e = 1,\ \lambda_e \geq -\beta} \sum_{e\in\mathcal{E}} \lambda_e R_e(\theta)$$



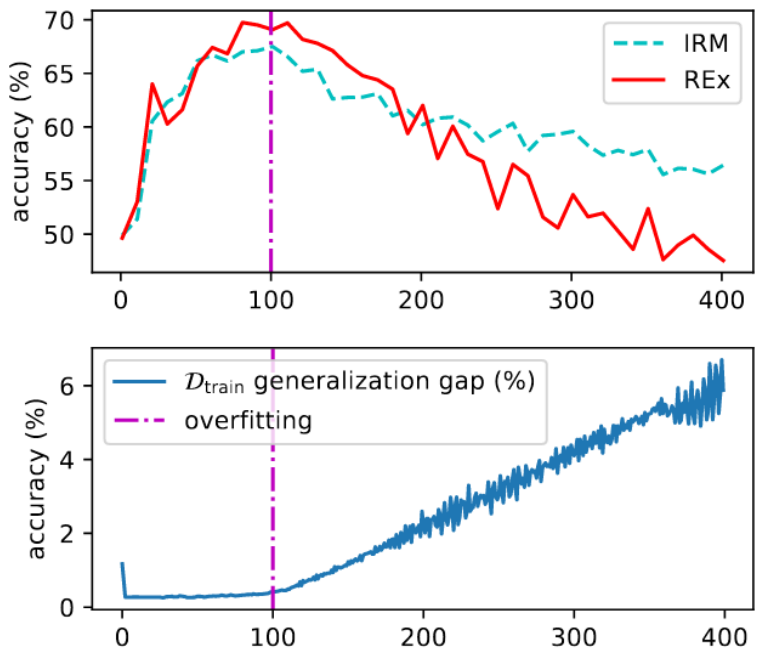
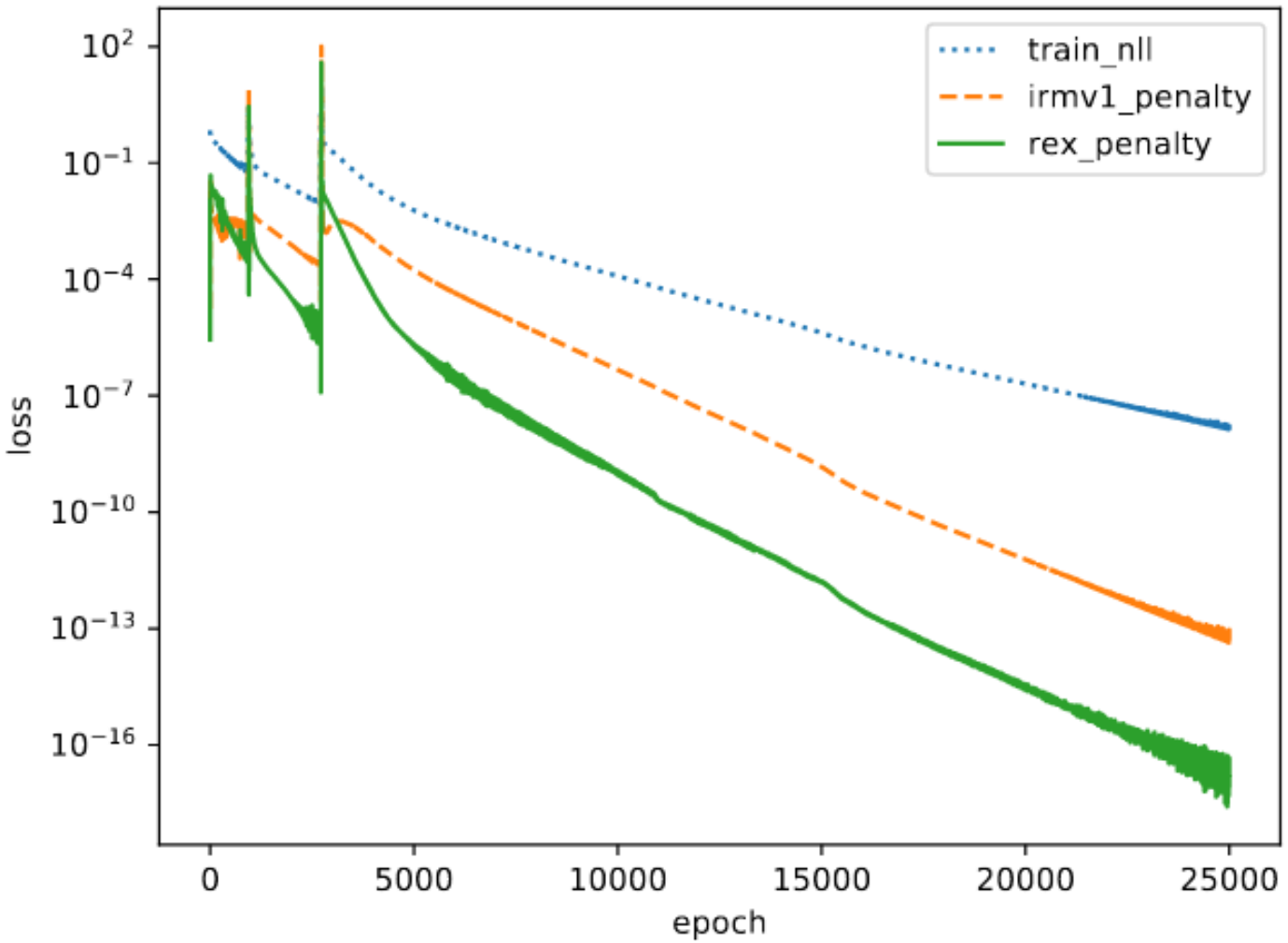
Hyperparameter Tuning?

The main challenge: To identify a suitable validation distribution.

Issue: One should not have access to any distribution that is very close to the test distribution. On the other hand, it is also important to have access to a validation distribution that is sufficiently different from the training distribution.

How to learn stable features?

Successful learning of stable features should have two stages. In the first stage, predictive features are learned. In the second stage, stable features are selected and/or predictive features are fine-tuned for stability.



Empirical Risk Minimization